

REPORT

MACHINE LEARNING

ASSIGNMENT 1

(CONCEPT LEARNING)

Group # 60

GROUP MEMBERS

ASRA ANEES

ALI ABDULLAH

I. INTRODUCTION

Nowadays with the increase in technology communication is a significant part of our life there are many ways of verbal or non-verbal communication. One way is email known as communication via the internet it can be used for various purposes i.e., in marketing, business, chatting, etc. As the growth of internet users increases the sending or receiving email has increased.

The unwanted junk emails are known as spam emails which became a big problem on the internet and may lead to computer network security risks. It wastes a lot of time to read spam emails and delete them. So, it's a best practice to have a spam filter. Spam filters detect junk or unwanted email and protect our inboxes from receiving spam emails.

The dataset "spambase.data" that is given to us, after analysis, is continuous data. And we need to scale and transformed it into discrete data before giving it to the model. Then we used the process of binning to do that, which will increase the performance of the model and decrease the number of feature values.

We used the Least general generalization (LGG) unsupervised learning algorithm to train our model. The aim of the algorithm is to identify the most specific classification that accurately categorizes all the instances in a given dataset.

II. PROCESS FOLLOWED

A. Dataset:

The spam base dataset contains 58 columns and 8 rows we apply the concept learning technique and find the hypostasis size and LGG (Least General Generalization) rule. There are three types of LGG rules.

1. LGG -Set – find LGG of the set of instances.
2. LGG-Conj(x,y) - find LGG of two conjunctions.
3. LGG-Conj-ID(x,y) – find the LGG of two conjunctions with internal disjunction.

In this report, we focused on rule 1 & 2.

B. Pre-processing

The function 'preprocess_data' takes in a data frame as input and performs two main pre-processing steps on the data. First, it scales the continuous features in the dataset using StandardScaler, which standardizes the features by subtracting the mean and dividing by the standard deviation. This helps to improve the model's performance by ensuring that all features have equal importance in the analysis.

Secondly, the function uses KBinsDiscretizer to transform the continuous features into discrete values by binning. This reduces the number of possible values for each feature and can simplify the model's computations. The function returns the transformed features (X) and the target variable (y), which can then be used to train and test a machine-learning model.

C. Training and testing:

The methods that we used for training our model are "LGG" and "LGGConjunction". These procedures are applied to train the model by repeatedly finding the most general concept that can capture the common features of the data in the training set. These methods are used to find the LGG of data concepts that can represent a set of data.

The function 'train_model' trains a machine-learning model using two algorithms, LGG (z) and LGGConjunction(x,y). The function returns the result of the LGG algorithm applied to the training data (X_train), which represents the most general concept that captures the common features of the data. This result can be used to make predictions on new data and evaluate the model's performance.

Declaring 80% of positive class spam data as training data and for the test, we take the rest of 20 % of positive class and the whole negative class ham data.

We then test the model by using the 'test_model' function that evaluates the performance of a trained model on a test set by making predictions on the test set's features given as "X_test". For each feature in 'X_test', the function checks if the feature is contained in the most general concept referred to as "m", computed during training using the LGG algorithm. If the feature is contained in the most general concept, the function predicts that it belongs to the same group as the training data. Otherwise, it predicts that the feature belongs to a different group. The function returns a list of predicted values that can be compared to the actual target variable given as "y_test" to assess the model's accuracy.

D. Model Creation

Create the model by creating 50 bins. And then discretizing the feature of training and testing data into equal 8-width bins. take out a unique feature of each. Using discretized data calculate the size of hypothesis space which is equal to $2^{(50+1)^{57}}$. It also defines the conjunction concept as $(50+1)^{57}$ for the given number of bins.

III. PERFORMANCE MEASURE

Calculate the accuracy, precision, recall, specificity, FPR(False positive rate), FNR (False Negative Rate), and F1-score by following formulas:

A. Accuracy

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

B. PRECISION

$$Precision = TP / (TP + FP)$$

C. RECALL

$$Recall = TP / (TP + FN)$$

D. SPECIFICITY

$$Specificity = TN / (TN + FP)$$

E. FALSE POSITIVE RATE

$$false\ positive\ rate = FP / (FP + TN)$$

F. FALSE NEGATIVE RATE

$$\text{false negative rate} = FN/(FN + TP)$$

G. F1 SCORE

$$f1 - Score = 2 * (Recall * Precision) / (Recall + Precision)$$

The code defines a function named "compute_metrics" that takes in two arguments, the true labels and the predicted labels. It then calculates several performance metrics such as accuracy, precision, recall, specificity, false positive rate, false negative rate, and F1 score using functions from the sci-kit-learn library such as "accuracy_score", "precision_score", "recall_score", "confusion_matrix", and "f1_score". It uses tuple unpacking to assign the values returned by "confusion_matrix" to variables named true_negative, false_positive, false_negative, and true_positive. Finally, the function returns all the computed metrics as a tuple.

IV. RESULT

The the information about size of bin, hypothesis space, and conjunctions:

[illegible]

The result of all classifications is:

```
-----Confusion matrix-----
ACCURACY:          57.655 %
PRECISION:         0.0 %
RECALL:            0.0 %
SPECIFICITY:       100.0 %
FALSE POSITIVE RATE: 0.0 %
FALSE NEGATIVE RATE: 100.0 %
```

V. CONCLUSION

This report presents the classification of emails classify them into ham and spam mail. For this purpose, we used the best technique to classify two objects of spam base dataset LGG-Conjunction rule.